

RESEARCH STUDY - FEBRUARY 2026

AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors:

US/Europe Divergence Trajectories 2024-2030

Integrated Geostrategic and Economic Analysis - ANNEX C - ACADEMIC NOTE

**76.9% of global operational AI
compute = USA**

**1.59x energy cost EU/US (PPP-
adjusted)**

**3.46:1 CACI US/EU ratio (Power
Mode)**

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ANNEX C - ACADEMIC NOTE

AI for Americans First: American AI Protectionism, Reconfiguration of the Global Technological Order and Consequences for France and Europe (2026-2030)

This annex presents the Academic Synthesis Note of the dissertation. This study analyzes the mechanisms and consequences of American AI protectionism under the Trump 2.0 administration, integrating four dimensions: energy, semiconductors, compute, and regulation. Based on an empirical diagnosis (2020-2026), the construction of a Compute-Adjusted Competitiveness Index (CACI), and a 2x2 scenario matrix, the research demonstrates that the combination of tariffs (25%, Section 232) and export controls creates a measurable structural competitive advantage (CACI US/EU ratio of 3.46:1, Power mode, for a raw operational compute ratio of 17.6:1, April 2026). The analysis reveals differentiated trajectories of dependence across regions (Europe, South America, Asia, Africa). A prospective addendum formalizes the 'Grand Decoupling 2028' scenario based on US Cloud Sovereignty Mandates, revealing that France and Europe must distinguish between physically installed capacity and operationally sovereign compute.

C.1 Subject and Problem Statement

AI has emerged since 2023 as the primary vector of economic innovation and geopolitical competition. Yet the AI value chain exhibits unprecedented concentration: as of the April 2026 public dashboard snapshot, the United States controls 76.9 percent of global operational AI compute, and five US hyperscalers plan 660-690 billion USD in capex for 2026 alone.

In this context, the Trump 2.0 administration has transformed Biden-era export controls (2022-2025) into a hybrid protectionist regime. This study asks: to what extent does American AI protectionism create a measurable structural competitive advantage, and what are the differentiated consequences for France, Europe, and other world regions?

C.2 Methodological Framework

The methodology rests on three pillars: (1) An empirical longitudinal diagnosis (2020-2026) of energy, semiconductors, and compute distributions; (2) The construction of the Compute -Adjusted Competitiveness Index (CACI) Power Mode formula: $CACI = F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$; (3) A 2x2 scenario matrix crossing US protectionism intensity with European strategic response.

A prospective addendum extends the CACI model with the decomposition $F(r) = F_{phys}(r) \times F_{sov}(r)$, where F_{sov} measures the fraction of regional compute operated outside US jurisdiction. This reveals that physically installed capacity does not equal operationally sovereign capacity.

C.3.1 A Three-Tiered Protectionism

The study identifies a cumulative three-level protectionist architecture: (1) Export controls segmenting the world into three access tiers; (2) 25% Section 232 tariffs on advanced AI semiconductors creating a direct cost differential; (3) Capitalistic gravity effects where massive capex concentration reinforces US supremacy without further regulation.

A potential fourth tier looms for 2028: Cloud Sovereignty Mandates, transforming US hyperscalers operating offshore into conditional intermediaries of global compute.

C.3.4 The Grand Decoupling 2028

This scenario models a qualitative shift: the transition from control of chip flows to control of the jurisdictional layer (operating sovereignty). The CLOUD Act (2018) already establishes that a cluster physically installed in Ireland remains legally American. The 'America's AI Action Plan' (2025) introduces location-verification features in chips for remote throttling.

Decomposition reveals that while the US and China are fully sovereign ($F_{sov}=1.00$), the EU is largely sovereign on cluster ownership ($F_{sov}=0.99$) but highly dependent on the cloud workload layer ($F_{sov_workloads} \sim 0.28$).

Table C.1. Summary of regional consequences of American AI protectionism.

Source: Author's construction, calibrated on April 2026 snapshot.

Region	Tier	Main Dynamic	Strategic Asset	Main Risk
France / Europe	1	Dependence on US	Nuclear energy (70%)	Geopolitical vendor

		GPU + cloud; InvestAI response	mix), Mistral, ASML, AI Act	lock-in; Low F_sov; CSM 2028
Brazil / S. America	2	US-China competition arena; Scala/TikTok projects	83% renewable energy mix; dynamic fintech market	Triple fracture (N-S, E-W, intra-regional)
China	3	Forced autonomy; alternative ecosystem (Huawei/DeepSeek)	Domestic market 1.4B; 125B+ USD/yr investment	2-3 gen GPU lag; tech isolation

Main sources: IEA (2025-2026), World Bank (2025), Epoch AI, McKinsey, Synergy Research, White House BIS.

Notes

1. Public dashboard snapshot April 2026: USA 76.9% of global operational AI compute.
2. CACI Power Mode: $F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$. Validated at $\beta=0.251$, $p<0.01$.
3. Section 232: 25% tariff on advanced AI semiconductors (Jan 15, 2026).
4. CLOUD Act (2018): Jurisdictional control over data regardless of physical storage location.

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Public dashboard: <https://moOgly.github.io/America-First-IA/dashboard/>

Repository: <https://github.com/moOgly/America-First-IA>

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